

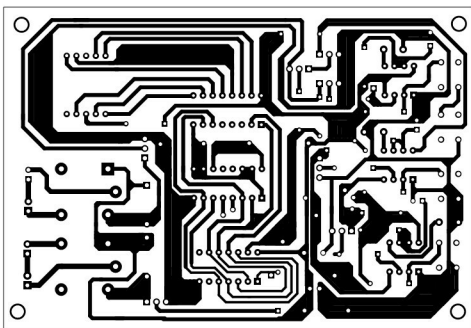


Fig. 2: An actual-size, single-side PCB for the automatic room-lock and lights circuit

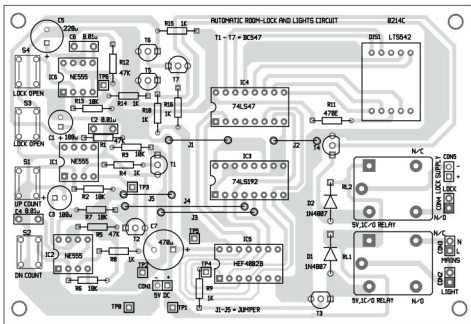


Fig. 3: Component layout for the PCB

makes the base of transistor T4 low, via transistor T5. This action de-energises relay RL2 to open the lock for a pre-defined time period based on the values of R12 and C5. At least one person should enter the room within this time period, otherwise it will get locked again. During this period, Vcc pin 8 remains high to enable IC1.

When each person enters the room and presses foot switch S1, counter IC3

advances by one count and display shows the number of persons in the room. On first person's entry, the outputs O1 and O2 of IC5 go low, transistors T3 and T4 stop conducting and relays RL1 and RL2 de-energise to switch on the light and open the door lock.

Similarly, when they start leaving the room by enabling foot switch S2, counter IC3 reduces by one count and the display shows the number of

PARTS LIST

Semiconductors:	
IC1, IC2, IC6	- NE555 timer
IC3	- 74LS192 synchronous up / down decade counter
IC4	- 74LS47, 7-segment decoder / driver
IC5	- HEF4002B dual 4-input NOR gate
T1-T7	- BC547 npn transistor
DIS1	- L18542 common anode 7-segment display
D1-D2	- 1N4007 rectifier diode
Resistors (all 1/4-watt, 5% carbon):	
R1, R5, R12	- 47-kilo-ohm
R2, R3, R6, R7, R13	- 10-kilo-ohm
R4, R8-R10, R14-R16	- 1-kilo-ohm
R11	- 470-ohm
Capacitors:	
C1, C3	- 100µF, 16V electrolytic
C2, C4, C6	- 0.01µF ceramic disk
C5	- 220µF, 16V electrolytic
C7	- 470µF, 16V electrolytic
Miscellaneous:	
CON1-CON5	- 2-pin connector
S1, S2	- Foot switch
S3, S4	- Push-to-on switch
RL1, RL2	- 5V, 1 C/O relay
	- 5V DC adaptor
	- Door-lock assembly

persons present in the room. The outputs O1 and O2 of IC5 remain low, so light remains on and door remains open till the room is vacant.

Construction and testing

An actual-size, single-side PCB for the circuit is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on PCB, enclose it in a suitable case. Switches S1 and S2 should be installed as explained earlier. The display can be installed on the door frame outside the room. Fix switches S3 and S4 at a suitable location such that you can press them simultaneously when needed.

Connect the door-lock assembly to relay RL2 contacts using external wires as shown in Fig. 1. Use a 2-pin connector each for connecting the light and lock assembly.

EFY note. Avoid entry of more than nine persons. Otherwise, at the entry of 10th person, the display will show '0,' the door will get locked and the lights will switch off. If this happens, someone will have to press switch S2 from inside the room to open the lock and switch on the lights. ●

The author is a lab engineer at EFY Lab